13.9. NATURAL RADIOACTIVITY OF THE OCEANS

The total amounts of ^{238}U and ^{232}Th in the ocean are 4.3×10^{12} kg (53 EBq) and 6.9×10^{10} kg (0.3 EBq), respectively. The oceans contain a much higher concentration of uranium than would be expected from its abundance in rocks. It also occurs at much larger concentration than thorium, even though the latter element is more abundant in rocks. This is explained by the absence of easily soluble thorium complexes, in contrast to the strong $UO_2(CO_3)_3^{4-}$ complex. ^{238}U decays to ^{234}Th ($t_{½}$ 24.1 d), which precipitates out or absorbs to colloids before it decays to ^{234}U, causing a slight disequilibrium between the two U-isotopes. This is used to determine the mean residence time of Th in the oceans: ~200 y; by contrast, the residence time of U in the ocean is 5×10^5 y. The decay passes the long-lived ^{230}Th ($t_{½}$ 75 400 y), which deposits in sediments before radioactive equilibrium is reached with ^{226}Ra (t 1600 y); consequently the specific activity of ^{226}Ra is less than expected from the U-content (39 Bq/m^3), only 1–10 Bq/m^3. It should be noted that the ^{222}Rn concentration in surface waters is in disequilibrium with ^{226}Ra because Rn is vented by heat, turbulence, etc.

As the total content of ^{40}K in the oceans is 7.4×10^{13} kg, corresponding to 1.94×10^4 EBq, ^{40}K is the largest source of radioactivity in the oceans. Minor activities come from ^3H as HTO and from ^{14}C as dissolved CO_2 or HCO_3^-.

13.10. ANTHROPOGENIC RADIOACTIVITY IN NATURE

In the analysis of a sample for its content of natural radioactivity it is necessary today to consider the possibility that the sample has become contaminated by "non-natural" radioactivities, i.e. radionuclides added to by human activities (so-called *anthropogenic sources*). Nuclear weapons tests, nuclear satellites burnt-up in the atmosphere, and nuclear power accidents may release large amounts of activities, see Table 13.4. The nuclear power industry is permitted by health authorities to continually release small, controlled amounts of specified radionuclides into the atmosphere and into open waters, Tables 19.7 and 21.10. Effluents may also come from large radiotracer users (especially

Table 13.4 Events leading to large injections of radionuclides into the atmosphere

Source	Country	Time	Radioactivity (Bq)	Important nuclides
Hiroshima & Nagasaki	Japan	1945	4×10^{16}	Fiss.prod., Actinides
Atmospheric weapons tests	USA USSR	-1963	2×10^{20}	Fiss. prod., Actinides
Windscale	UK	1957	1×10^{15}	^{131}I
Chelyabinsk (Kysthym)	USSR	1957	8×10^{16}	Fiss. prod. ^{90}Sr, ^{137}Cs
Harrisburg	USA	1979	1×10^{12}	Noble gases, ^{131}I
Chernobyl	USSR	1986	2×10^{18}	^{137}Cs
Fukushima	Japan	2010	2×10^{18}	^{131}I, ^{137}Cs